Ass 5 chat

Client Side Code

```
import socket

s = socket.socket()
s.connect(("127.0.0.1", 5000))

msg = input("Client: ")
s.send(msg.encode())

reply = s.recv(1024).decode()
print("Server:", reply)

s.close()
```

Server Side Code

```
import socket

s = socket.socket()
s.bind(("127.0.0.1", 5000))
s.listen(1)

print("Waiting for client...")

d, addr = s.accept()
print("Client connected")

msg = d.recv(1024).decode()
print("Client:", msg)

reply = input("Server: ")
d.send(reply.encode())

d.close()
s.close()
```

Arithmetic

Client Side Code

```
import socket

s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
s.connect(("127.0.0.1", 5000))

a = input("Enter first number: ")
op = input("Enter operation (+, -, /): ")
b = input("Enter second number: ")

msg = a + " " + op + " " + b
s.send(msg.encode())

result = s.recv(1024).decode()
print("Result:", result)

s.close()
```

Server Side Code

```
import socket

s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
s.bind(("127.0.0.1", 5000))
s.listen(1)

print("Waiting for client...")
c, addr = s.accept()
print("Client connected")

data = c.recv(1024).decode()

a, op, b = data.split()
a = float(a)
b = float(b)

if op == "+":
    result = a + b
elif op == "-":
    result = a - b
elif op == "*":
    result = a * b
elif op == "/":
    result = a / b
else:
    result = "Invalid operator"

c.send(str(result).encode())
c.close()
```